

OpenRNG

Patent Portfolio



COMPUTATIONAL TRUST INFRASTRUCTURE

5-PATENT FAMILY

July 2, 2026

OpenRNG · Taipei, Taiwan

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SECTION I

Executive Summary

Thesis

Computational trust is a new infrastructure layer. Every random number has a trust problem — traditional systems return a value and discard the process that created it. OpenRNG introduces **Verifiable Entropy** — transforming randomness from a disposable output into a portable, auditable, and independently verifiable digital object.

The Five Patents — Logical Curve

#	TECHNICAL PROBLEM	SYSTEM PROPERTY	STATUS	PATENTABILITY
1	Random values cannot be individually verified	Verifiable Randomness	✅ Approved	—
2	Random values can be previewed before commitment	Dual Temporal Integrity	📄 Ready to file	High
3	Trust disappears after randomness is consumed	Portable Trust Assertion	📄 Ready to file	High
4	Computational trust cannot survive delegation	End-to-End Verifiable Trust	🔬 Problem defined	70–80%
5	Trust quality silently degrades during operation	Continuous Trust Health	🔬 Problem defined	80–90%

The Logical Progression

#1 Can randomness be verified? → #2 Can randomness be hidden until it should exist? → #3 Can trust survive after randomness is consumed? → #4 Can trust survive delegation? → #5 Can trust survive time?

Strategic Value

- **Patents #1–#3:** Gaming / regulated industry foundation
- **Patent #4:** AI agent delegation (OpenAI, multi-agent systems)
- **Patent #5:** Highest commercial value — "Trust Health" as new category (like Battery Health)
- **Big Tech interest:** OpenAI → #4/#5 · Google/Microsoft → #5 (governance) · Gaming → #1–#3
- **Strategic decision:** Stop after #5 — next patent emerges from real usage

Independent Review Scores

PATENT	PATENTABILITY	COMMERCIAL VALUE	DEFENSIBILITY	OVERALL
#1 Verifiable Randomness	9/10	7/10	7/10	7.7
#2 Dual Temporal Integrity	8/10	8/10	8/10	8.0
#3 Portable Trust Assertion	7/10	7/10	6/10	6.7
#4 End-to-End Verifiable Trust	6/10	8/10	5/10	6.3
#5 Continuous Trust Health	8/10	9/10	7/10	8.0

Portfolio-level assessment: **7.3/10** — Strong, coherent portfolio with genuine commercial potential.

SECTION I-A

Portfolio Risk Matrix

Consolidated risk assessment across the five-patent family, based on independent review findings. Each risk is scored by potential impact and current likelihood, with recommended mitigations.

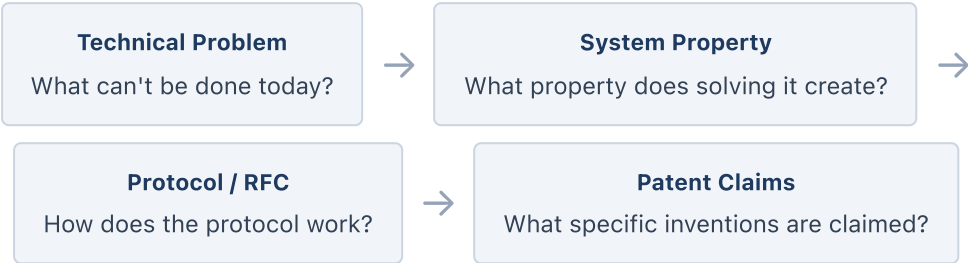
RISK	IMPACT	LIKELIHOOD	MITIGATION
Chainlink claim overlap (US12401532B2)	HIGH	MEDIUM	Claim-by-claim analysis needed — obtain granted claim language and conduct formal FTO comparison
Alice §101 rejection (#4, #5)	HIGH	MEDIUM	Anchor all independent claims to hardware VDF computational bottleneck; maintain specific data structures in claims
Cyberscope blocking patent (Aug 2025)	MEDIUM	LOW	Monitor prosecution; differentiate on entropy-specific trust vs. smart contract audit focus
Prior art combination — §103 obviousness	MEDIUM	MEDIUM	Dual-path pipeline (sub-2ms serving + slow anchor) as key non-obvious engineering tradeoff defense
Priority date delay (#2, #3)	HIGH	HIGH	File #2 and #3 provisionals immediately — every week of delay is exposure
Self-inflicted prior art (open-source repo)	MEDIUM	MEDIUM	Audit public OpenRNG repository for pre-filing disclosures
VDF algorithm cryptographic break	HIGH	LOW	Ensure claims are algorithm-agnostic where possible; monitor cryptographic research

Highest-priority action: File Patents #2 and #3 immediately to establish priority dates. The combination of priority date delay risk (HIGH/HIGH) and Chainlink claim uncertainty makes this the most time-sensitive item in the portfolio.

SECTION II

Framework

Problem → Property → Protocol → Patent



Dual Naming Convention

LAYER	PURPOSE	AUDIENCE
Technical Problem language	Precise problem definition	Patent examiners, engineers, attorneys
System Property language	Capability description	Papers, talks, standards, industry
Movement language	Vision & narrative	Manifestos, vision ("The Second Law of Computational Trust")

Rule: Never mix these layers.

PATENT #1

Verifiable Randomness

Technical Problem: Random values cannot be independently verified

System Property: Verifiable Randomness

Status:  **APPROVED**

Title: Method and System for Gaming Random Number Generation

PATENTABILITY

9/10

COMMERCIAL VALUE

7/10

DEFENSIBILITY

7/10

The Problem

When a casino pays out \$2 million on a jackpot, regulators need to verify the winning number was genuinely random. Current systems produce audit logs written by the same system they audit. The randomness itself carries no proof.

The Invention

A system that generates random numbers as **Verifiable Entropy Objects (VEOs)** — cryptographic containers carrying entropy payload, source provenance, confidence metrics, provider signatures, and blockchain anchors. Any party can independently verify a VEO without trusting the provider.

Core Innovation

The **VEO-1 standard** — transforms randomness from a disposable value into a verifiable digital object with five layers: Payload, Provenance, Confidence, Proof, and Anchor.

Key Technical Components

- **VDF** (Verifiable Delay Functions) for time-locked entropy generation
- **drand** public randomness beacon for independent entropy sourcing
- **Merkle tree batching** for efficient verification of large token pools
- **Polygon blockchain anchoring** for immutable proof
- **Entropy Confidence Score (ECS):** 0–1000 quality metric across 6 dimensions

⚠ RISK ASSESSMENT

Approved and foundational — strongest legal footing in the portfolio. However, Chainlink's US12401532B2 disclosure explicitly mentions VDF combination. While their *claims* are specific to distributed VRF oracle networks, the disclosure creates prior art exposure that requires a formal claim-by-claim analysis to quantify. The centralized-vs-distributed architectural distinction is the primary moat.

PATENT #2

Dual Temporal Integrity

Technical Problem: Random values can be previewed before commitment

System Property: Dual Temporal Integrity

Status:  **Ready to file (Provisional + Non-Provisional drafted)**

PATENTABILITY

8/10

COMMERCIAL VALUE

8/10

DEFENSIBILITY

9/10

The Problem

The **"Insider Preview Attack"** — in traditional RNG, the system generating randomness can pre-view values before they're committed. An operator, insider, or compromised system could select favorable outcomes.

The Invention

A dual-phase temporal commitment scheme where:

1. Random values are generated and cryptographically committed (Merkle root anchored on-chain) **BEFORE** any request arrives
2. Values are revealed only upon request, with the pre-commitment proving they existed before the request

Core Innovation

Separating the generation timeline from the consumption timeline through VDF-enforced time delays + blockchain anchoring, making preview attacks mathematically impossible.

Differentiation from Chainlink VRF

Chainlink uses request-response (randomness generated AFTER request). OpenRNG pre-generates pools of committed values — fundamentally different temporal model.

⚠ RISK ASSESSMENT

Cleanest claims in the portfolio. Strongest §103 defense via the dual-path pipeline — pre-generation temporal model is fundamentally different from Chainlink's request-response architecture. The pre-generated pool with blockchain-anchored Merkle roots is structurally distinct from every existing verifiable randomness system. Examiner will likely cite general commitment schemes (Blum 1981); defense must emphasize the pool architecture, not the commitment concept.

PATENT #3

Portable Trust Assertion

Technical Problem: Trust disappears after randomness is consumed

System Property: Portable Trust Assertion

Status:  **Ready to file**
(Provisional + Non-Provisional drafted)

PATENTABILITY

7/10

COMMERCIAL VALUE

7/10

DEFENSIBILITY

7/10

The Problem

Once a random number is consumed, the trust evidence that backed it is discarded. A slot machine uses a random number, records the outcome, but the cryptographic proof of that number's integrity is lost. Trust is disposable.

The Invention

Trust Assertions that persist after entropy consumption — portable, verifiable records that any party can independently verify at any future time. The VEO becomes a permanent trust artifact, not a disposable value.

Core Innovation

The concept of "**entropy receipts**" — each consumption generates a portable attestation record that carries full provenance, quality scores, and verification paths. Trust becomes a first-class, persistent data object.

Key Continuation Anchors

This patent's non-provisional specification includes the Five-Layer Trust Stack and Decision Attestation Records — securing earliest priority dates for Patents #4 and #5.

⚠ RISK ASSESSMENT

Solid but carries the broadest claims of the gaming cluster. The "entropy receipt" concept is novel but must be aggressively distinguished from standard digital receipts, certificates, and blockchain-based proof-of-existence systems. A hostile examiner could characterize this as "logging with cryptographic signatures." The Five-Layer Trust Stack and ECS quality metrics are the key differentiators — without them, the claims collapse into existing attestation/notarization prior art.

PATENT #4

End-to-End Verifiable Trust

Technical Problem:	Computational trust cannot survive delegation	System Property:	End-to-End Verifiable Trust
Status:	 Problem defined, drafts completed	Patentability:	70–80%

PATENTABILITY	COMMERCIAL VALUE	DEFENSIBILITY
6 _{/10}	8 _{/10}	5 _{/10}

The Problem

When a trusted computation delegates work to another agent, the trust breaks. The delegation boundary destroys three things:

1. **Provenance linkage** — delegated agent gets task but not trust artifact
2. **Quality continuity** — no mechanism to detect trust quality degradation through delegation
3. **Audit continuity** — auditor cannot trace back to original trust assertion

Why Existing Approaches Fail

- **Task queues** (Celery, SQS) deliver tasks, not trust
- **API auth** (OAuth, JWT) proves WHO delegated, not WHY trustworthy
- **Distributed tracing** (OpenTelemetry) traces execution path, not trust path
- **Blockchain logs** record THAT delegation occurred, not the trust that AUTHORIZED it
- **PKI** proves identity delegation, not computational trust delegation

The Invention

A system for preserving computational trust across delegation boundaries by cryptographically linking child trust assertions to parent assertions through lineage records that enforce non-amplification, record degradation transparently, and enable independent verification of the complete trust chain from any node to root — without cooperation from intermediate nodes.

Target Market

Multi-agent AI systems (OpenAI, Anthropic), cross-organizational computation, regulatory compliance chains.

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⚠ **RISK ASSESSMENT**

Weakest link in the portfolio. Independent claims are currently too narrow — over-specified to DAG mechanics, making the patent easy to design around. Needs an abstraction pass to broaden independent claims while keeping DAG as one embodiment. Carries the highest Alice §101 risk: "preserving trust across delegation boundaries" sounds abstract. **Central tension:** broadening claims for defensibility increases §101 exposure. The DAG specificity is actually a §101 safe harbor — losing it requires careful reanchoring to concrete technical operations.

Action Needed: *Add broader independent claim beyond DAG mechanics — let DAG be one embodiment, not the only claim.*

PATENT #5

Continuous Trust Health

Technical Problem:	Trust quality silently degrades during operation	System Property:	Continuous Trust Health
Status:	 Problem defined, drafts completed	Patentability:	80–90% (highest of all five)

PATENTABILITY	COMMERCIAL VALUE	DEFENSIBILITY
8/10	9/10	7/10

The Problem — Trust Decay

Trust infrastructure doesn't fail catastrophically — it rots. Entropy sources silently degrade. Quality scores drift. Trust chains grow deeper than policy allows. Anchoring latency creeps up. Source diversity narrows. None triggers an error. Each is individually small. Collectively, they transform a AAA-grade system into LOW-grade — without anyone noticing.

The Battery Analogy

Trust Decay is to trust infrastructure what battery drain is to mobile devices. A fully charged battery (AAA trust) slowly loses charge. The drain is not an error — it is physics. Without monitoring, the user doesn't know until the device dies. Trust infrastructure needs the same: a trust health meter, decay detection, and active maintenance.

Why Existing Tools Fail

- **SIEM** (Splunk) monitors security events, not trust quality
- **APM** (Datadog) monitors performance — system is fast while trust silently degrades
- **GRC platforms** require manual periodic checks — Trust Decay is continuous
- **Alerting systems** catch threshold breaches — Trust Decay is sub-threshold (many small drifts)

The Invention

A system for detecting, quantifying, and counteracting continuous trust decay through:

- Multi-dimensional trust health monitoring
- Anomaly detection against trust-specific metrics (not security/performance)

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- Automated corrective response (source rebalancing, policy tightening)
- Continuous machine-readable compliance attestation
- Predictive trust quality analytics forecasting decay before critical thresholds

Commercial Value

Highest of all five patents. "Trust Health" creates a new monitoring category. Every compliance-heavy industry needs this. Google/Microsoft interested for governance/compliance.

RISK ASSESSMENT

Highest novelty and commercial value in the portfolio. "Trust Health" as a new category is a strong framing — no existing patent or academic paper treats trust quality as a time-varying signal requiring continuous monitoring. Key threats: Cyberscope's Aug 2025 patent filing for "AI-Optimized Blockchain Trust Scoring Platform" and Yaslioglu's 2025 arXiv paper on Bayesian time-decay trust in blockchain. Both are distinguishable (Cyberscope = smart contract audit; Yaslioglu = reputation trust in P2P networks), but expect examiners to cite them.

SECTION III

Whitepaper: OpenRNG — The Verifiable Entropy Network

Core Concept

Verifiable Entropy Objects (VEO-1) — cryptographic containers for randomness carrying payload, provenance, confidence, proof, and anchor layers.

Object Classes

CLASS	NAME	DESCRIPTION
VEO-1A	Raw Source	Single-source entropy
VEO-1B	Composite	Multi-source aggregated
VEO-1C	Anchored	Composite + blockchain anchor
VEO-1D	Decision	Verifiable Decision Objects (reserved)

Entropy Confidence Score (ECS)

0–1000 quality metric measuring freshness, source diversity, independence, manipulation resistance, and verification success.

"Randomness is a value. Entropy is infrastructure." — Like electricity has meters, grids, and regulatory infrastructure, randomness needs the same.

SECTION IV

Prior Art Landscape

HIGH RISK**1. US12401532B2 — Chainlink (GRANTED)**

Distributed VRF with blockchain verification. Mentions VDF in disclosure.

Differentiation: Chainlink = distributed oracle + VRF (secret key). OpenRNG = centralized server + VDF (time-based) + Merkle batching. Different architecture, different primitive, different trust model.

2. US10291408B2 — Amazon (GRANTED)

Merkle tree generation from PRNG values.

Differentiation: Amazon = PoW mining challenges. OpenRNG = random token verification. Different use case entirely.

MEDIUM RISK**3. US20220410017A1 — nChain (ABANDONED)**

Provably fair gaming + blockchain.

Differentiation: nChain = interactive commit-reveal. OpenRNG = non-interactive pre-generated batches.

4. WO2017053754A1 — Spur Trail (CEASED)

Generic provably fair gaming. No Merkle batching, no VDF.

Key Differentiation

OpenRNG's sub-2ms in-memory token injection + slow background Merkle proof anchoring (dual-path pipeline: Security vs Performance) is the primary defense against §103 obviousness.

SECTION IV-A

New Threats

Emerging filings and publications identified during independent review that may affect prosecution or create prior art exposure.

◆ Cyberscope — AI-Optimized Blockchain Trust Scoring (August 2025)

Filing: US patent application for "AI-Optimized Blockchain Trust Scoring Platform"

Relevance: The "trust scoring + blockchain" combination creates potential §103 combination prior art, particularly against Patent #5 (Continuous Trust Health). While likely focused on smart contract auditing rather than entropy trust, the overlap in language and domain requires monitoring.

Impact: **MEDIUM** **Affected patents:** #4, #5

Action: Monitor prosecution; prepare differentiation brief anchored to entropy-specific trust monitoring vs. smart contract audit scoring.

◆ Yaslioglu (2025) — Bayesian-Time-Decay Trust Mechanism in Blockchain

Publication: arXiv preprint — "A Novel Bayesian-Time-Decay Trust Mechanism in Blockchain"

Relevance: Directly addresses time-decay of trust in blockchain contexts using Bayesian methods. An examiner will almost certainly cite this paper against Patent #5. The specific mechanisms differ significantly (Bayesian reputation decay in P2P networks vs. OpenRNG's multi-dimensional computational trust quality monitoring), but the surface-level language overlap is high.

Impact: **MEDIUM** **Affected patents:** #5

Defense: Yaslioglu addresses *reputation* trust decay (node trustworthiness based on past behavior). OpenRNG addresses *computational* trust quality decay (cryptographic and statistical properties of generated entropy). Different type of trust, different decay mechanism, different monitoring approach, different remediation.

Monitoring cadence: Both threats should be tracked quarterly. Cyberscope prosecution documents will become available through USPTO PAIR. Yaslioglu's paper should be checked for journal publication or citation in other patent filings.

SECTION V

Red Team Summary

The three highest-probability attack vectors opposing counsel would deploy against this portfolio, and the consolidated defense playbook.

Top 3 Attack Vectors

Attack #1 — §103 Obviousness Combination

Target: Patents #1, #2, #3

Argument: "Chainlink VRF (US12401532B2) + Amazon Merkle trees (US10291408B2) = obvious combination. VDF is known (Boneh 2018). Blockchain anchoring is known (Bitcoin 2008). The combination of these known techniques to verify randomness is obvious to one skilled in the art."

Threat level: **HIGH** — This is the most likely attack and will be attempted during prosecution.

Attack #2 — Alice Corp §101 Abstract Idea

Target: Patents #4, #5

Argument: "Trust management is an abstract concept. 'Preserving trust across delegation boundaries' (#4) and 'monitoring trust quality over time' (#5) are mental processes that humans perform. Implementing them on a computer does not transform them into patent-eligible subject matter."

Threat level: **HIGH** — §101 rejections are increasingly common for software patents. Patent #4 is most exposed.

Attack #3 — §102 Anticipation via Academic Prior Art

Target: Patents #1, #2

Argument: "VDF academic papers (Boneh et al. 2018) + commit-reveal schemes (Blum 1981) anticipate the core claims. The specific combination of pre-generation with VDF-derived seeds was foreseeable from the academic literature."

Threat level: **MEDIUM** — Anticipation requires a single reference; the combination argument is weaker here than §103.

Defense Playbook

✓ PRIMARY DEFENSE: HARDWARE VDF COMPUTATIONAL BOTTLENECK

- VDF computation is inherently sequential — cannot be parallelized or shortcut
- This hardware constraint is the §101 safe harbor for #4/#5: the claims are anchored to a physical computational limitation, not an abstract idea
- No human can perform VDF computation, Merkle tree verification, or real-time entropy source rebalancing — these are inherently computational processes

✓ SECONDARY DEFENSE: DUAL-PATH PIPELINE

- Sub-2ms in-memory token serving + slow background Merkle anchoring = non-obvious engineering tradeoff
- No prior art system separates the serving path (in-memory, instant) from the verification path (blockchain, delayed)
- This architectural choice is not suggested by any combination of Chainlink + Amazon + academic papers

✓ TERTIARY DEFENSE: PRE-GENERATION POOL ARCHITECTURE

- Pre-generation pool model is architecturally distinct from all prior art request-response models
- Chainlink generates randomness AFTER request; OpenRNG generates BEFORE — fundamentally different temporal architecture
- Batches of 65,536 VDF-derived tokens committed as a single Merkle root — no existing system proposes this

SECTION VI

Filing Status & Next Steps

Filing Status

PATENT	DOCUMENT	STATUS
#1	Original Patent	✓ Approved
#2	Provisional	✓ Drafted
#2	Non-Provisional	✓ Drafted
#3	Provisional	✓ Drafted
#3	Non-Provisional	✓ Drafted
#4	Problem Definition	✓ Complete
#4	Provisional Draft	✓ Drafted
#4	Non-Provisional Draft	✓ Drafted
#5	Problem Definition	✓ Complete
#5	Provisional Draft	✓ Drafted
#5	Non-Provisional Draft	✓ Drafted

Filing Strategy & Priority Order

Priority order: #2 → #3 → #5 → #4

18-Month Budget Breakdown

ITEM	ESTIMATED COST
Patent #2 — Provisional + Non-Provisional	\$8K – \$15K
Patent #3 — Provisional + Non-Provisional	\$8K – \$15K
Patent #4 — Provisional + Non-Provisional	\$10K – \$18K
Patent #5 — Provisional + Non-Provisional	\$10K – \$18K
PCT applications for #2 and #3	\$12K – \$20K
Attorney review + prosecution (office actions)	\$9K – \$10K
Total 18-month plan	\$57K – \$96K

Immediate Next Steps

1. **File Patents #2 and #3 provisionals immediately** — ready now, every week of delay is risk
2. **Obtain and analyze Chainlink US12401532B2 granted claims** — claim-by-claim comparison is the single biggest unknown
3. **Insert Five-Layer Trust Stack + Decision Attestation** in Patent #3 non-provisional (secures priority for #4/#5)
4. **Patent #4: Abstraction pass** — broaden independent claim beyond DAG mechanics while maintaining §101 compliance
5. **Patent #5: Maintain "Trust Health" framing** — crown jewel, deserves most careful drafting
6. **Audit public OpenRNG repository** for pre-filing disclosures that could constitute self-inflicted prior art
7. **Strategic decision confirmed:** Stop after #5, let reality drive #6

Smart Contract

0xA79E149C35Ad47Ed270Bf4b16B80170eBF7B88F8 (Polygon Amoy)